PROFESSIONAL TRAINING

AWS Certified DevOps Engineer Professional Training

TGS-2025054815

Tertiary Infotech Academy Pte Ltd
UEN: 201200696W
Trainer: Dr. Alfred Ang | v1

ADMINISTRATION

Digital Attendance (TRAQOM · SSG)

01 Scan the TRAQOM QR code shown by the trainer

02 Record morning and afternoon attendance

03 Use the same registered learner details

04 Tell the trainer immediately if attendance fails

About the Trainer

?

Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation and software engineering.

DELIVERS

WSQ courses on cloud, DevOps, automation and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

WELCOME

Learner Introduction

01 **Your role and organisation**

02 **Your AWS and DevOps experience**

03 **One CI/CD challenge you want to solve**

04 **Your target exam date**

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively - no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

COURSE ROADMAP

Two-day learning progression

1

Day 1: governance, IAM, infrastructure as code and CI/CD

2

Day 2: deployments, operations, observability, resilience, security and exam review

3

Final 2 hours: written assessment and practical performance

LEARNING OUTCOMES

What you will be able to do

01 **Automate secure SDLC workflows**

02 **Implement infrastructure as code and detect drift**

03 **Select safe deployment strategies**

04 **Operate fleets with Systems Manager**

05 **Design observability and incident response**

06 **Improve resilience, security and cost controls**

Briefing for Assessment

01 Assessment is open book

02 Complete Written Assessment before Practical Performance

03 Follow the assessor's timing and evidence rules

04 Submit answers through the LMS

05 Ask process questions before assessment starts

06 Sign the Assessment Summary Record

Assessment and Funding

1

Written Assessment: 1 hour

2

Practical Performance: 1 hour

3

Competency requires attendance and successful assessment evidence

Assessment Flow Reminder

1. **TRAQOM digital attendance**
2. **Assessment digital attendance**
3. **Sit WA then PP**
4. **Submit answers on the LMS**
5. **Sign Assessment Summary Record**

The TRAQOM QR code and assessment submission are both handled through the LMS/TMS portal.

Access the Hands-On Labs

1

https://github.com/tertiarycourses/TGS-2025054815---AWS-Certified-DevOps-Engineer-Professional-Training/tree/main/labs

2

Option A: git clone the course repository

3

Option B: GitHub Code > Download ZIP

AWS Workbench and Cost Safety

01 Use the trainer-provided AWS account where available

02 Choose the instructed AWS Region

03 Never expose access keys or secrets

04 Create budgets and monitor charges

05 Prefer free-tier eligible resources

06 Delete paid resources after each lab

DOMAIN

SDLC Automation and Infrastructure as Code

01

Build capability through labs 1, 2, 3.

HANDS-ON LAB 1

Account Safety, IAM, DevOps Tooling, and Governance Baseline

Scenario

Professional-level DevOps work starts with account safety, auditability, and controlled access. You will review the foundation used by CI/CD, automation, monitoring, and operations labs.

Evidence to retain

- IAM and least-privilege notes
- Governance and audit notes
- DevOps service mapping table
- Budget or cost-safety notes

LAB 1

Objectives and decision points

01 Establish a safe AWS DevOps learning baseline.

02 Review IAM, roles, permission boundaries, and least privilege.

03 Review governance, audit, and cost-safety services.

04 Map core DevOps services to SDLC activities.

Follow the lab sequence and capture evidence

1. **Sign in to the AWS account assigned by your trainer.**
2. Confirm the account and region.
3. Open IAM.
4. Verify MFA where permitted.
5. Review users, groups, roles, policies, and permission boundaries.
6. Review how service roles are used by CodeBuild, CodePipeline, CloudFormation, and Systems Manager.

LAB 1 CHECKPOINT

Verify the outcome before moving on

✓

- **You should be able to explain the governance baseline needed before automating AWS delivery and operations.**

HANDS-ON LAB 2

Infrastructure as Code with CloudFormation, CDK, Config, and Drift

Scenario

Infrastructure as code supports repeatable environments and controlled change. You will review how AWS tracks desired state, detects drift, and evaluates configuration compliance.

Evidence to retain

- CloudFormation feature notes
- CDK and StackSets notes
- AWS Config compliance notes
- IaC governance workflow

LAB 2

Objectives and decision points

01 Review CloudFormation stacks, templates, change sets, and drift detection.

02 Understand CDK and StackSets at a high level.

03 Review AWS Config rules and compliance snapshots.

04 Design an IaC governance workflow.

Follow the lab sequence and capture evidence

1. **Open CloudFormation.**
2. Review stacks, templates, parameters, outputs, mappings, conditions, and resources.
3. Review change set behavior.
4. Review stack rollback and termination protection concepts.
5. Review drift detection.
6. Create a stack only if your trainer permits it.

LAB 2 CHECKPOINT

Verify the outcome before moving on

✓

- **You should be able to explain how IaC, drift detection, and Config support controlled AWS operations.**

CI/CD with CodePipeline, CodeBuild, CodeDeploy, and Artifact Stores

Scenario

DevOps engineers automate safe application delivery. This lab designs a CI/CD workflow with testing, artifact management, approvals, deployment, and rollback.

Evidence to retain

- CI/CD service comparison table
- Pipeline architecture
- IAM role notes
- Rollback and notification notes

Objectives and decision points

01 Design a CI/CD pipeline.

02 Review buildspec, artifacts, test stages, approvals, and deployment stages.

03 Compare CodePipeline, CodeBuild, CodeDeploy, and CodeArtifact concepts.

04 Plan rollback and failure handling.

Follow the lab sequence and capture evidence

1. **Open CodePipeline.**
2. Review pipeline stages, actions, transitions, and approvals.
3. Open CodeBuild.
4. Review build projects, build environments, buildspec files, artifacts, reports, and logs.
5. Open CodeDeploy.
6. Review applications, deployment groups, deployment configurations, lifecycle hooks, and rollback.

LAB 3 CHECKPOINT

Verify the outcome before moving on

✓

- You should be able to explain professional CI/CD design, including artifacts, approvals, validation, and rollback.

DOMAIN

Deployment, Configuration and Operations

02

Build capability through labs 4, 5.

Deployment Strategies for Lambda, ECS, EC2, and Elastic Beanstalk

Scenario

Different workloads require different deployment strategies. You will compare controlled release patterns and choose safe deployment approaches.

Evidence to retain

- Deployment strategy matrix
- Health check and rollback notes
- Traffic shifting notes
- Monitoring requirements

LAB 4

Objectives and decision points

01 Compare deployment strategies for different AWS compute platforms.

02 Review blue/green, canary, linear, rolling, immutable, and all-at-once deployments.

03 Understand traffic shifting, health checks, and rollback.

04 Design a deployment plan for multiple workload types.

Follow the lab sequence and capture evidence

1. **Review Lambda versions and aliases.**
2. Review Lambda traffic shifting with CodeDeploy concepts.
3. Review ECS services, task definitions, deployment controllers, and rolling updates.
4. Review blue/green deployments for ECS using CodeDeploy at a high level.
5. Review EC2 Auto Scaling rolling or immutable deployment concepts.
6. Review Elastic Beanstalk deployment policies.

LAB 4 CHECKPOINT

Verify the outcome before moving on

✓

• You should be able to select a deployment strategy based on workload type, risk, rollback needs, and availability requirements.

Systems Manager, Automation, Patch, Parameter Store, and Run Command

Scenario

DevOps teams operate fleets and automate routine administration. This lab reviews AWS Systems Manager as a control plane for secure operations.

Evidence to retain

- Systems Manager capability table
- Parameter Store vs Secrets Manager comparison
- Patching workflow
- Automation runbook outline

Objectives and decision points

01 Review Systems Manager operations capabilities.

02 Understand Run Command, Session Manager, Automation, Patch Manager, and State Manager.

03 Compare Parameter Store and Secrets Manager.

04 Design fleet operations and remediation workflows.

Follow the lab sequence and capture evidence

1. **Open Systems Manager.**
2. Review managed nodes and prerequisites.
3. Review Run Command.
4. Review Session Manager and why it can reduce SSH exposure.
5. Review Automation runbooks.
6. Review Patch Manager.

LAB 5 CHECKPOINT

Verify the outcome before moving on

✓

- You should be able to explain how Systems Manager supports secure fleet operations and automation.

DOMAIN

Monitoring, Incident Response and Resilience

03

Build capability through labs 6, 7.

HANDS-ON LAB 6

Observability with CloudWatch, X-Ray, CloudTrail, EventBridge, and Alarms

Scenario

Reliable DevOps requires visibility into system behavior. You will design a monitoring and incident signal model for a distributed application.

Evidence to retain

- Observability design
- Alarm and SLI list
- Log retention notes
- Event-driven incident routing notes

Objectives and decision points

01 **Design observability for AWS applications and infrastructure.**

02 **Use metrics, logs, traces, dashboards, and alarms conceptually.**

03 **Review X-Ray and distributed tracing.**

04 **Route operational events with EventBridge.**

Follow the lab sequence and capture evidence

1. **Open CloudWatch.**
2. Review metrics, namespaces, dimensions, dashboards, and alarms.
3. Review CloudWatch Logs, log groups, metric filters, and subscription filters.
4. Review anomaly detection and composite alarms at a high level.
5. Open X-Ray.
6. Review service maps, traces, segments, subsegments, and sampling.

Verify the outcome before moving on

✓

- You should be able to design observability that supports troubleshooting, alerting, and automated response.

HANDS-ON LAB 7

High Availability, Backup, Disaster Recovery, and Auto Remediation

Scenario

Professional DevOps engineers design for failure. This lab covers availability, backup, routing, scaling, and automated recovery.

Evidence to retain

- HA and DR pattern comparison
- Backup and restore plan
- RTO/RPO notes
- Auto remediation workflow

Objectives and decision points

01 Review high availability and disaster recovery patterns.

02 Understand backup and restore services.

03 Design auto remediation workflows.

04 Compare RTO, RPO, and resilience tradeoffs.

LAB 7 PROCEDURE

Follow the lab sequence and capture evidence

1. **Review Multi-AZ and multi-region concepts.**
2. Review Auto Scaling groups and scaling policies.
3. Review Elastic Load Balancing health checks.
4. Review Route 53 routing policies and health checks.
5. Open AWS Backup and review backup plans, vaults, retention, and restore concepts.
6. Review RDS backup, snapshots, Multi-AZ, and read replica concepts.

LAB 7 CHECKPOINT

Verify the outcome before moving on

✓

- You should be able to explain resilience design choices and automated recovery patterns for AWS applications.

DOMAIN

Security, Compliance, Cost and Exam Readiness

04

Build capability through labs 8.

Security, Compliance, Cost Optimization, and Exam Review

Scenario

The final lab connects security, compliance, cost management, governance, and exam readiness for AWS DevOps Engineer Professional scenarios.

Evidence to retain

- Security and compliance checklist
- Cost optimization checklist
- Exam-domain revision map
- Timed practice score

Objectives and decision points

01 Review security and compliance guardrails for DevOps.

02 Plan cost optimization for delivery and operations.

03 Map labs to exam domains.

04 Complete timed certification-style practice.

Follow the lab sequence and capture evidence

1. **Open Security Hub and review security posture concepts.**
2. Open GuardDuty and review threat detection concepts.
3. Review IAM Access Analyzer and policy validation concepts.
4. Review KMS, Secrets Manager, and Parameter Store.
5. Review AWS Config rules and conformance packs.
6. Review Service Control Policies and permission boundaries at a high level.

Verify the outcome before moving on

✓

- **You should be ready to reason through advanced AWS DevOps scenarios covering delivery, operations, resilience, security,...**

COURSE RECAP

What You Achieved

01 Governed AWS accounts and IAM

02 Designed repeatable infrastructure

03 Mapped CI/CD and deployment controls

04 Planned fleet automation and patching

05 Connected metrics, logs, traces and events

06 Applied resilience, security and cost guardrails

NEXT STEPS

Courseware and Assessment on the LMS

1

Download the PPT and Learner Guide

2

Complete the assessment on the LMS

3

LMS: https://lms-tms.tertiaryinfotech.com/

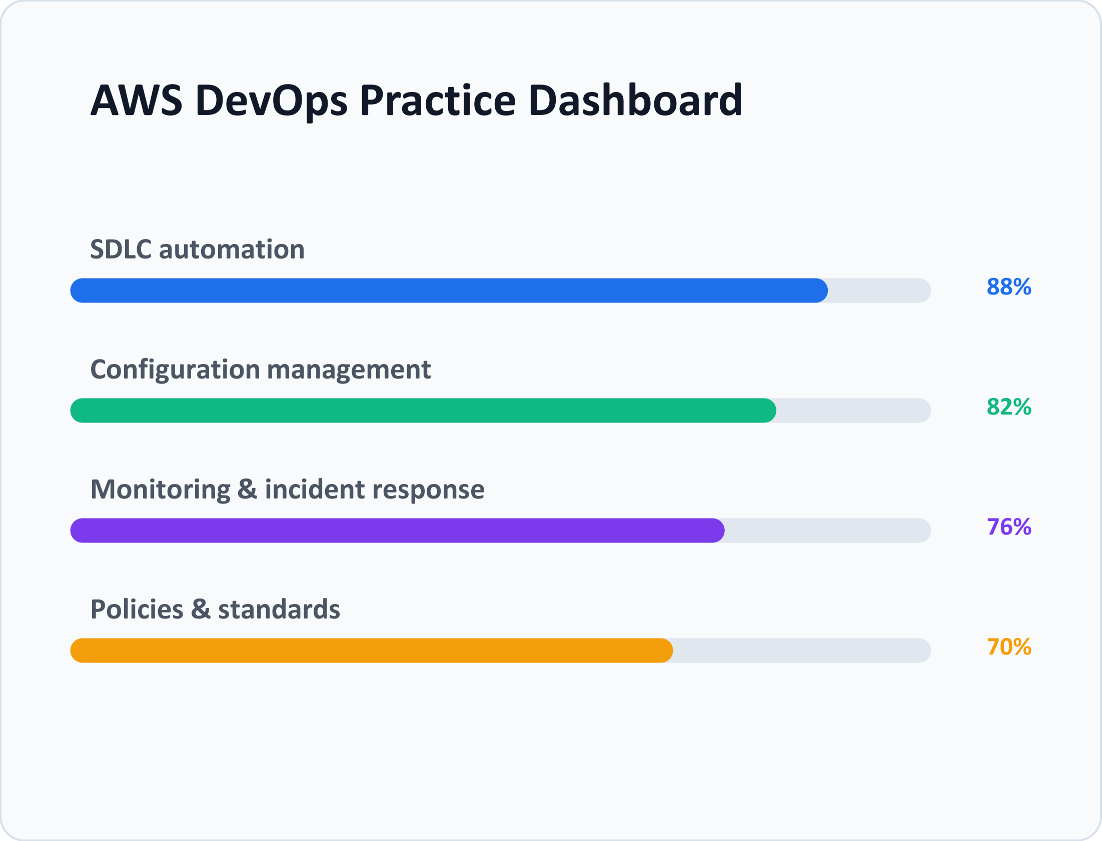
PRACTICE EXAM

Prepare with AWS DevOps practice questions

- Reinforce all AWS DevOps Engineer Professional domains with realistic exam-style questions.
- Use timed sets to test SDLC automation, configuration management, monitoring, incident response, security and governance.
- Review incorrect answers and revisit the linked lab evidence.

Access and practise at:

AWS Skill Builder or trainer-provided practice set



ASSESSMENT

Final Assessment Reminder

1

Written Assessment - 1 hour

2

Practical Performance - 1 hour

3

Open book; submit on LMS and retain evidence

Assessment Flow Reminder

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MANDATORY

Digital Attendance (TRAQOM · SSG)

01 Complete the assessment attendance record

02 Use your registered learner details

03 Confirm successful submission

04 Inform the assessor if the record fails

Thank You

Apply the lab evidence to your AWS DevOps practice and exam preparation.

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